Year 8

REVISION PACK

Unit 1

ANSWERS

Teacher copy: worked answers to every practice question in the revision pack. Working is set out one line under another, as students are expected to show it.

Unit 1 · Negative numbers

Q	Working	Answer
calculating with negative numbers		
1.1	**(a)** $-14 - 22 = -36$ **(b)** $-6 \times 12 = -72$ **(c)** $84 \div -7 = -12$ **(d)** $8 - (-13) = 8 + 13 = 21$	**(a) -36** **(b) -72** **(c) -12** **(d) No. Minus a minus is plus: 8 + 13 = 21**
1.2	**(a)** $-17 - 9 = -26$ **(b)** $-8 \times 11 = -88$ **(c)** $-72 \div 8 = -9$ **(d)** $5 - (-9) = 5 + 9 = 14$	**(a) -26** **(b) -88** **(c) -9** **(d) No. Minus a minus is plus: 5 + 9 = 14**
1.3	**(a)** $-25 - 16 = -41$ **(b)** $7 \times -9 = -63$ **(c)** $96 \div -12 = -8$ **(d)** $3 - (-14) = 3 + 14 = 17$	**(a) -41** **(b) -63** **(c) -8** **(d) No. Minus a minus is plus: 3 + 14 = 17**
1.4	**(a)** $-11 - 33 = -44$ **(b)** $-12 \times 5 = -60$ **(c)** $-66 \div -6 = 11$ **(d)** $9 - (-4) = 9 + 4 = 13$	**(a) -44** **(b) -60** **(c) 11** **(d) No. Minus a minus is plus: 9 + 4 = 13**
1.5	**(a)** $-28 - 7 = -35$ **(b)** $13 \times -4 = -52$ **(c)** $132 \div -11 = -12$ **(d)** $6 - (-7) = 6 + 7 = 13$	**(a) -35** **(b) -52** **(c) -12** **(d) No. Minus a minus is plus: 6 + 7 = 13**

1.6	**(a)** $-19 - 19 = -38$ **(b)** $-15 \times 6 = -90$ **(c)** $-150 \div 10 = -15$ **(d)** $10 - (-15) = 10 + 15 = 25$	**(a) -38** **(b) -90** **(c) -15** **(d) No. Minus a minus is plus: 10 + 15 = 25**

Unit 1 · Prime factors, HCF and LCM

Q	Working	Answer
prime factors, HCF and LCM		
2.1	**(a)** $540 \div 2 = 270,\ 270 \div 2 = 135,\ 135 \div 3 = 45,\ 45 \div 3 = 15,\ 15 \div 3 = 5,\ 5 \div 5 = 1$ **(b)** smaller powers: $2^2 \times 3 = 12$ **(c)** bigger powers: $2^2 \times 3^3 \times 5 \times 7 = 3780$	**(a)** $2^2 \times 3^3 \times 5$ **(b) 12** **(c) 3780**
2.2	**(a)** $360 \div 2 = 180,\ 180 \div 2 = 90,\ 90 \div 2 = 45,\ 45 \div 3 = 15,\ 15 \div 3 = 5,\ 5 \div 5 = 1$ **(b)** smaller powers: $2^2 \times 3 = 12$ **(c)** bigger powers: $2^3 \times 3^2 \times 5 \times 7 = 2520$	**(a)** $2^3 \times 3^2 \times 5$ **(b) 12** **(c) 2520**
2.3	**(a)** $250 \div 2 = 125,\ 125 \div 5 = 25,\ 25 \div 5 = 5,\ 5 \div 5 = 1$ **(b)** smaller powers: $2 \times 5 = 10$ **(c)** bigger powers: $2^2 \times 3 \times 5^3 = 1500$	**(a)** 2×5^3 **(b) 10** **(c) 1500**
2.4	**(a)** $144 \div 2 = 72,\ 72 \div 2 = 36,\ 36 \div 2 = 18,\ 18 \div 2 = 9,\ 9 \div 3 = 3,\ 3 \div 3 = 1$ **(b)** smaller powers: $2^4 \times 3 = 48$ **(c)** bigger powers: $2^5 \times 3^2 = 288$	**(a)** $2^4 \times 3^2$ **(b) 48** **(c) 288**
2.5	**(a)** $420 \div 2 = 210,\ 210 \div 2 = 105,\ 105 \div 3 = 35,\ 35 \div 5 = 7,\ 7 \div 7 = 1$ **(b)** smaller powers: $2^2 \times 5 \times 7 = 140$ **(c)** bigger powers: $2^2 \times 3 \times 5 \times 7 = 420$	**(a)** $2^2 \times 3 \times 5 \times 7$ **(b) 140** **(c) 420**

2.6	**(a)** $80 \div 2 = 40,\ 40 \div 2 = 20,\ 20 \div 2 = 10,\ 10 \div 2 = 5,\ 5 \div 5 = 1$ **(b)** smaller powers: $2^2 \times 5 = 20$ **(c)** bigger powers: $2^4 \times 3 \times 5^2 = 1200$	**(a)** $2^4 \times 5$ **(b) 20** **(c) 1200**

Unit 1 · Order of operations

Q	Working	Answer
powers, roots and order of operations		
3.1	**(a)** $16 + 36 \div 4$ $= 16 + 9 = 25$ **(b)** $(16 + 36) \div 4$ $= 52 \div 4 = 13$ **(c)** $81 + 2 \times 5$ $= 81 + 10 = 91$ **(d)** $(-2) \times (-2) = 4$ $4 \times (-2) = -8$	**(a) 25** **(b) 13** **(c) 91** **(d) shown**
3.2	**(a)** $9 + 81 \div 9$ $= 9 + 9 = 18$ **(b)** $(9 + 81) \div 9$ $= 90 \div 9 = 10$ **(c)** $16 + 5 \times 4$ $= 16 + 20 = 36$ **(d)** $(-5) \times (-5) = 25$ $25 \times (-5) = -125$	**(a) 18** **(b) 10** **(c) 36** **(d) shown**
3.3	**(a)** $64 + 16 \div 4$ $= 64 + 4 = 68$ **(b)** $(64 + 16) \div 4$ $= 80 \div 4 = 20$ **(c)** $81 + 4 \times 2$ $= 81 + 8 = 89$ **(d)** $(-6) \times (-6) = 36$ $36 \times (-6) = -216$	**(a) 68** **(b) 20** **(c) 89** **(d) shown**

3.4	**(a)** $36 + 64 \div 4$ $= 36 + 16 = 52$ **(b)** $(36 + 64) \div 4$ $= 100 \div 4 = 25$ **(c)** $16 + 10 \times 3$ $= 16 + 30 = 46$ **(d)** $(-7) \times (-7) = 49$ $49 \times (-7) = -343$	**(a) 52** **(b) 25** **(c) 46** **(d) shown**
3.5	**(a)** $81 + 144 \div 9$ $= 81 + 16 = 97$ **(b)** $(81 + 144) \div 9$ $= 225 \div 9 = 25$ **(c)** $256 + 3 \times 10$ $= 256 + 30 = 286$ **(d)** $(-1) \times (-1) = 1$ $1 \times (-1) = -1$	**(a) 97** **(b) 25** **(c) 286** **(d) shown**
3.6	**(a)** $4 + 100 \div 4$ $= 4 + 25 = 29$ **(b)** $(4 + 100) \div 4$ $= 104 \div 4 = 26$ **(c)** $625 + 6 \times 4$ $= 625 + 24 = 649$ **(d)** $(-10) \times (-10) = 100$ $100 \times (-10) = -1000$	**(a) 29** **(b) 26** **(c) 649** **(d) shown**

Unit 1 · Laws of indices

Q	Working	Answer
multiplying, dividing and powers of powers		
4.1	**(a)** $5 + 4 = 9$ **(b)** $8 - 2 = 6$ **(c)** $3 \times 5 = 15$	**(a)** 8^9 **(b)** 8^6 **(c)** 8^{15}

4.2	**(a)** $4 + 6 = 10$ **(b)** $10 - 4 = 6$ **(c)** $5 \times 3 = 15$	**(a)** 5^{10} **(b)** 3^{6} **(c)** 2^{15}
4.3	**(a)** $2 + 8 = 10$ **(b)** $12 - 5 = 7$ **(c)** $3 \times 6 = 18$	**(a)** 9^{10} **(b)** 8^{7} **(c)** 4^{18}
4.4	**(a)** $5 + 5 = 10$ **(b)** $9 - 1 = 8$ **(c)** $4 \times 2 = 8$	**(a)** 3^{10} **(b)** 2^{8} **(c)** 10^{8}
4.5	**(a)** $6 + 3 = 9$ **(b)** $11 - 7 = 4$ **(c)** $2 \times 8 = 16$	**(a)** 4^{9} **(b)** 5^{4} **(c)** 3^{16}
4.6	**(a)** $8 + 7 = 15$ **(b)** $6 - 2 = 4$ **(c)** $6 \times 3 = 18$	**(a)** 2^{15} **(b)** 9^{4} **(c)** 5^{18}

Unit 1 · Using the laws of indices

Q	Working	Answer
showing results and writing as a power of 2		
5.1	**(a)** $3^{3-2} \times 5 = 3^{1} \times 5$ $= 3 \times 5 = 15$ **(b)** $(5^{1})^{3} = 5^{3} = 125$ **(c)** $\dfrac{2^{4} \times 2^{5}}{2^{5}} = 2^{4+5-5}$ **(d)** $2^{7} \div 2^{3} = 2^{7-3}$	**(a) shown** **(b) shown** **(c)** 2^{4} **(d)** 2^{4}

5.2	**(a)** $2^{6-4} \times 5 = 2^2 \times 5$ $= 4 \times 5 = 20$ **(b)** $(4^1)^3 = 4^3 = 64$ **(c)** $\frac{2^2 \times 2^5}{2^3} = 2^{2+5-3}$ **(d)** $2^5 \div 2^3 = 2^{5-3}$	**(a) shown** **(b) shown** **(c)** 2^4 **(d)** 2^2
5.3	**(a)** $5^{4-2} \times 3 = 5^2 \times 3$ $= 25 \times 3 = 75$ **(b)** $(7^1)^2 = 7^2 = 49$ **(c)** $\frac{2^5 \times 2^2}{2^4} = 2^{5+2-4}$ **(d)** $2^7 \div 2^3 = 2^{7-3}$	**(a) shown** **(b) shown** **(c)** 2^3 **(d)** 2^4
5.4	**(a)** $3^{6-3} \times 2 = 3^3 \times 2$ $= 27 \times 2 = 54$ **(b)** $(2^3)^2 = 2^6 = 64$ **(c)** $\frac{2^3 \times 2^6}{2^5} = 2^{3+6-5}$ **(d)** $2^4 \div 2^3 = 2^{4-3}$	**(a) shown** **(b) shown** **(c)** 2^4 **(d)** 2^1
5.5	**(a)** $2^{9-6} \times 9 = 2^3 \times 9$ $= 8 \times 9 = 72$ **(b)** $(10^2)^2 = 10^4 = 10000$ **(c)** $\frac{2^6 \times 2^1}{2^7} = 2^{6+1-7}$ **(d)** $2^6 \div 2^3 = 2^{6-3}$	**(a) shown** **(b) shown** **(c)** 2^0 **(d)** 2^3
5.6	**(a)** $4^{5-3} \times 5 = 4^2 \times 5$ $= 16 \times 5 = 80$ **(b)** $(6^1)^3 = 6^3 = 216$ **(c)** $\frac{2^4 \times 2^4}{2^2} = 2^{4+4-2}$ **(d)** $2^5 \div 2^3 = 2^{5-3}$	**(a) shown** **(b) shown** **(c)** 2^6 **(d)** 2^2

Unit 1 · Problems with LCM		
Q	**Working**	**Answer**
when things happen together again		
6.1	$28: 28, 56, 84, 112, 140$ $35: 35, 70, 105, 140$ $\text{LCM} = 140 \text{ minutes}$ $6:45 + 140 \text{ min} = 9:05$	**9:05 am**
6.2	$12: 12, 24, 36, 48, 60$ $20: 20, 40, 60$ $\text{LCM} = 60 \text{ minutes}$ $7:00 + 60 \text{ min} = 8:00$	**8:00 am**
6.3	$18: 18, 36, 54, 72$ $24: 24, 48, 72$ $\text{LCM} = 72 \text{ minutes}$ $2:00 + 72 \text{ min} = 3:12$	**3:12 pm**
6.4	$4: 4, 8, 12$ $6: 6, 12$ $\text{LCM} = 12 \text{ minutes}$ $9:30 + 12 \text{ min} = 9:42$	**9:42 am**
6.5	$25: 25, 50, 75, 100, 125, 150, 175, 200$ $40: 40, 80, 120, 160, 200$ $\text{LCM} = 200 \text{ minutes}$ $8:00 + 200 \text{ min} = 11:20$	**11:20 am**
6.6	$12: 12, 24, 36, 48$ $16: 16, 32, 48$ $\text{LCM} = 48 \text{ minutes}$ $10:00 + 48 \text{ min} = 10:48$	**10:48 am**